

RESEARCH PROBLEM DOCUMENT

The Spatial Generalisation Problem in Vision Transformers

*How Direct Patch-Position Assignment
Prevents True Spatial Understanding*

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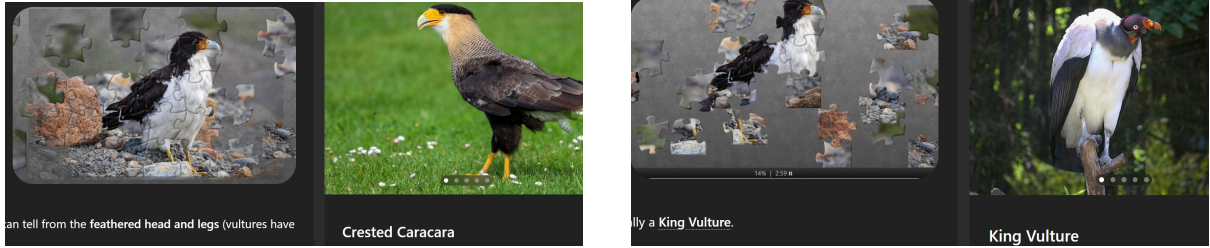
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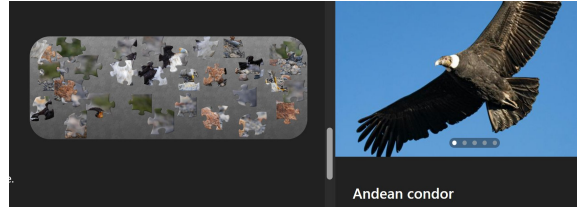
1 Motivating Observations: A Bird Identification Case Study

The screenshots below illustrate a practical scenario in which spatial context matters enormously for fine-grained visual recognition. Three birds — the **Crested Caracara**, the **King Vulture**, and the **Andean Condor** — are visually similar at coarse scales yet taxonomically distinct. A model that memorises fixed patch-position statistics will confuse them whenever the bird appears in an unexpected region or at an unusual scale.



(a) Crested Caracara — puzzle vs. reference.

(b) King Vulture — puzzle vs. reference.



(c) Andean Condor — scattered puzzle pieces vs. reference.

Figure 1: Bird identification under spatial disruption. Left column: puzzle representation (spatially disrupted patches). Right column: reference photograph with ground-truth label. A ViT using learned absolute positional embeddings struggles when patches appear in non-training positions.

The figures directly motivate the core thesis: *a model that binds recognition to fixed patch positions cannot generalise when those positions shift.*

2 Abstract

Vision Transformers (ViTs) have demonstrated impressive performance across visual recognition tasks, yet a persistent and fundamental limitation hampers their generalisation capability: the way spatial position is communicated to the model. The dominant approach — assigning each image patch a fixed, learned positional embedding — causes the model to *memorise* spatial locations rather than *learn* structural relationships. This leads to well-documented failures when the model encounters images at different resolutions, different aspect ratios, or with objects shifted from their training distribution.

This document formally defines the problem, traces its origins, surveys the landscape of proposed solutions, assesses honestly what remains unsolved, and positions a novel LLaMA-inspired architectural direction as a promising but under-explored research avenue.

3 The Core Problem: What Exactly Is Breaking?

3.1 How ViT Assigns Position

When an image is fed into a Vision Transformer, it is divided into a grid of fixed-size patches. Each patch is linearly projected into an embedding vector. To inject location awareness, Dosovitskiy et al. [2] add a *learned* positional embedding to each patch embedding:

$$\mathbf{x}_i = \text{patch}_i + \mathbf{p}_i \quad (1)$$

where $\mathbf{p}_i$ is a trainable vector, one per patch position. The critical detail: $\mathbf{p}_i$ is an **absolute address**. Patch at grid position (row = 3, col = 5) always receives the same embedding $\mathbf{p}_{(3,5)}$, regardless of context, scale, or content. The model has no mechanism to reason about *relative distances*.

Core Mechanism Failure. Absolute positional embeddings encode *spatial identity*, not *spatial relationships*. The model learns “position 42 is important here” rather than “this patch is 3 steps below that patch.” This kind of knowledge does not transfer across resolutions or layouts.

3.2 Why This Causes Non-Generalisation

3.2.1 Resolution Shift

A model trained on 224×224 images uses 196 patches at 16×16 . At test time on 384×384 images (576 patches), embeddings $\mathbf{p}_{197} \dots \mathbf{p}_{576}$ simply do not exist. Bilinear interpolation is a workaround, not a solution, and it degrades performance consistently.

3.2.2 Layout Memorisation

Objects commonly appear in specific training-distribution regions (faces centred, skies at top). The model exploits this positional bias as a shortcut; when objects appear in atypical positions at test time, performance drops even though the object is visually identical.

3.2.3 Aspect Ratio Sensitivity

Changing aspect ratio redistributes patches across the grid, creating a positional structure the model was never trained on. Embeddings no longer map to meaningful spatial locations.

3.2.4 Deep-Model Training Instability (Compounding Factor)

Original ViT uses Post-Norm (normalisation after residual addition). In very deep models this causes gradient vanishing, amplifying positional memorisation: the model converges to easy position-based shortcuts because the gradient signal for harder structural features is insufficient.

The Inductive Bias Gap. CNNs have translation invariance and locality built in by design. ViT has neither. Dosovitskiy et al. explicitly acknowledged this: ViT performs well only with very large-scale pre-training that compensates for the missing inductive bias through sheer data volume [2].

4 What the Literature Says: The Problem Is Confirmed

4.1 Foundational Paper

An Image is Worth 16x16 Words [2]. Without large-scale pre-training, ViT performs worse than CNNs — a deficit the authors attribute to the lack of CNN-like inductive biases. Learned absolute positional embeddings are a primary contributor.

4.2 Inductive Bias Analysis

On the Inductive Bias of Vision Transformers [9]. Systematically shows that ViT lacks locality and translation invariance. Learned positional embeddings become the primary (and insufficient) substitute for missing structural priors.

4.3 Local Attention as a Partial Fix

Swin Transformer [3]. Shifted-window attention and relative position bias within each window partially restore locality, but biases must still be interpolated for new resolutions.

4.4 Data-Efficient Training Highlights the Same Gap

DeiT [4]. ViT can train without massive datasets, but only with aggressive augmentation and distillation. The positional memorisation problem is managed, not solved.

4.5 Adding CNN Bias Helps

Early Convolutions Help Transformers See Better [10]. Replacing the patch embedding stem with a small CNN improves stability and accuracy — indirect confirmation that spatial structure from direct patching with absolute embeddings is insufficient.

5 Proposed Solutions: The Landscape

Table 1: Comparison of positional encoding methods in vision transformers.

Method	New Resolutions	Relative Awareness	Novelty	Status
Absolute Learned	No (interp. needed)	None	Baseline	Default ViT
Relative Bias	Partial	Within window	None	Widely adopted
Conditional (CPVT)	Yes	Limited	Moderate	Niche
2D RoPE	Yes	Full 2D	Moderate	Active research
Spiral RoPE	Yes	2D + diagonal	High	Very recent (2026)

5.1 Absolute Positional Embeddings (Baseline)

Simple lookup table; one trainable vector per patch. Works when train/test resolution matches; fails otherwise.

5.2 Relative Position Bias (Swin [3])

Adds learned biases to attention based on relative distance between patches. Better generalisation within trained resolution range; still resolution-dependent.

5.3 Conditional Positional Encodings — CPVT [11]

Generates encodings conditioned on input tokens, naturally handling variable input sizes. Not universally adopted due to added complexity.

5.4 Rotary Position Embeddings — RoPE [5]

Encodes position by rotating query and key vectors. The inner product decays naturally with distance and extrapolates to unseen lengths without interpolation. Standard in modern language models (LLaMA [6], Mistral, Gemma).

$$\mathbf{q}_m^\top \mathbf{k}_n = (R_\Theta^m \mathbf{q})^\top (R_\Theta^n \mathbf{k}) = \mathbf{q}^\top R_\Theta^{n-m} \mathbf{k} \quad (2)$$

The inner product depends only on the *relative offset* $n - m$. In the text domain this is 1-D; vision requires 2-D adaptation.

5.5 2D RoPE for Vision [7]

Extends RoPE to 2-D by applying rotations along x - and y -axes independently. Demonstrated improvements on ImageNet, COCO, and ADE20K. Not yet a framework-level default.

5.6 Spiral RoPE [8]

Adds multi-directional rotations (including diagonal) to address the axis-aligned limitation of basic 2D RoPE. Better spatial attention maps; consistently positive results. Most recent entry; not yet broadly validated.

6 Is There a Clear Breakthrough? Honest Assessment

Direct Answer: No. There is no single universally accepted breakthrough that completely solves the ViT positional generalisation problem. The field has not converged on one standard solution the way RoPE became standard for language models.

What has been solved:

- Absolute positional embeddings confirmed as the source of resolution sensitivity.
- Relative approaches (Swin, 2D RoPE) demonstrably improve generalisation.
- Trade-off between local and global attention is better understood.

What remains open:

- No canonical LLaMA-style ViT combining Pre-Norm + RMSNorm + RoPE as a unified package.
- 2D RoPE exists in isolated papers but is not the default in `timm` or `torchvision`.
- Spiral RoPE (2026) is promising but lacks broad experimental validation.
- Interaction between positional encoding and normalisation strategy in deep vision models has not been systematically studied.
- Whether LLaMA-style efficiency principles *compound* with better positional encoding in vision is an open question.

In text Transformers the story is clean: sinusoidal was replaced by RoPE, which is now dominant. Vision has no equivalent convergence yet. **This is the research gap.**

7 Formal Research Problem Statement

Primary Problem Statement. Vision Transformers assign spatial information to image patches through learned absolute positional embeddings — a mechanism that causes models to memorise fixed spatial layouts rather than learn generalisable structural relationships. This results in systematic failures at new resolutions, new aspect ratios, and shifted object distributions. Unlike CNNs, ViT has no built-in inductive bias for locality or translation invariance, making positional encoding the primary bottleneck for spatial generalisation.

7.1 Sub-Problem A: Resolution Non-Generalisation

Any deviation in image size requires interpolation of learned embeddings, introducing distortion and consistently degrading downstream performance.

7.2 Sub-Problem B: Layout Overfitting

Fixed location identity allows the model to exploit positional training-set patterns as shortcuts. Objects in atypical positions are harder to recognise even if visually identical.

7.3 Sub-Problem C: Absence of Relative Spatial Reasoning

Absolute encodings provide no mechanism to reason about spatial relationships between patches. The model knows *where things are*, but not *how they relate spatially*.

7.4 Sub-Problem D: Interaction with Depth and Normalisation

Post-Norm gradient instability compounds positional memorisation: unstable training causes convergence to easy position-based features because gradient signal for harder structural features is insufficient.

8 A Viable Research Direction

Research Question. Can LLaMA-style architectural design principles — specifically Pre-Layer Normalisation, RMSNorm, and Rotary Positional Embeddings extended to 2D — improve Vision Transformer spatial generalisation and training efficiency compared to the standard ViT baseline?

8.1 Motivation for LLaMA-Style Design

LLaMA [6] combined four changes over the original Transformer:

1. **Pre-Norm:** normalisation before each sub-layer, improving gradient flow.
2. **RMSNorm:** removes mean subtraction from LayerNorm, reducing compute while maintaining stability.
3. **SiLU activation:** $\text{SiLU}(x) = x \cdot \sigma(x)$, smooth non-linearity with better gradient flow than ReLU.
4. **RoPE:** replaces sinusoidal encodings with relative rotary embeddings.

The combination produced significant efficiency and scalability gains in language. Whether the same combination transfers to vision — and whether it specifically alleviates the spatial generalisation problem — has not been systematically studied.

8.2 RMSNorm

$$\text{RMS}(\mathbf{x}) = \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2}, \quad \text{RMSNorm}(\mathbf{x}) = \frac{\mathbf{x}}{\text{RMS}(\mathbf{x})} \cdot \gamma \quad (3)$$

Omits the mean-subtraction step of LayerNorm. Empirically yields $\approx 5\text{--}10\%$ compute reduction per normalisation call while preserving stability.

8.3 Pre-Norm vs. Post-Norm

The residual path under Pre-Norm is:

$$\mathbf{x}_{l+1} = \mathbf{x}_l + f(\text{Norm}(\mathbf{x}_l)) \quad (4)$$

Gradient flows directly through $\mathbf{x}_l$ without distortion, enabling very deep models. Post-Norm places normalisation after the residual, causing gradients to pass through an extra scaling operation at every layer, which accumulates to vanishing gradients in deep networks.

8.4 Proposed Experimental Ablation

Table 2: Proposed experimental variants for systematic ablation.

Variant	Norm Pos.	Norm Type	Pos. Encoding	Activation
Baseline ViT	Post	LayerNorm	Learned Absolute	GELU
PreNorm-ViT	Pre	LayerNorm	Learned Absolute	GELU
RMSNorm-ViT	Pre	RMSNorm	Learned Absolute	GELU
RoPE-ViT	Pre	RMSNorm	2D RoPE	GELU
LLaMA-ViT (full)	Pre	RMSNorm	2D / Spiral RoPE	SiLU

Evaluation metrics: top-1 accuracy (ImageNet-1K); resolution transfer (train 224^2 , test 384^2); convergence speed (epochs to 70% top-1); FLOPs per forward pass.

8.5 Novelty Positioning

The novelty lies in the **systematic combination and empirical characterisation** of whether these components interact positively in a vision setting. Individual papers have studied 2D RoPE alone [7] or Pre-Norm alone. The compound effect in a unified LLaMA-inspired vision architecture has not been characterised — that is the contribution.

9 Key References

References

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just normal assumption sir **One-Line Research Statement.** The absence of a canonical LLaMA-style Vision Transformer — combining Pre-Norm, RMSNorm, and 2D/Spiral RoPE as a unified architectural package — represents a clear and tractable research gap with direct implications for spatial generalisation, resolution transfer, and training efficiency in vision models.